

# CS1010: Discrete Mathematics for Computer Science

(Exam-1. Total: 30 marks.)

(Duration: 45 minutes. Date: 13 Oct 2025)

**Instructions:** ★ You may not get time to answer all the questions unless you have prepared really well. The exam is designed like that. ★ If your mobile phone is found with you during the exam, you will lose one grade. ★ Anybody found copying will get an F grade for the course straight away. ★ It is a no-break exam. You cannot take a break in between. The exam is only for 45 minutes. If you want to go out of the exam hall, you will have to submit your answer paper. ★ You should sit far apart from each other. The halls are big enough. If we see two students sitting close to each other, both the students will lose one grade.

## Questions

1. TRUE or FALSE (One word answer. No need to give any explanation). Assume  $a, b, c, n$  are positive integers.
  - (a) If  $\gcd(a, b) \neq 1$  and  $\gcd(b, c) \neq 1$ , then  $\gcd(a, c) \neq 1$ .
  - (b) If  $a|bc$  and  $\gcd(a, b) = 1$ , then  $a|c$ .
  - (c)  $\gcd(a^n, b^n) = (\gcd(a, b))^n$ .
  - (d)  $\gcd(ab, ac) = a \cdot \gcd(b, c)$ .
  - (e)  $\gcd(1 + a, 1 + b) = 1 + \gcd(a, b)$ .
  - (f) If an integer linear combination of  $a$  and  $b$  equals 1, then so does some integer linear combination of  $a$  and  $b^2$ .
  - (g) If no integer linear combination of  $a$  and  $b$  equals 2, then neither does any integer linear combination of  $a^2$  and  $b^2$ .
  - (h) One **cannot** obtain the integer 5 as an integer linear combination of 256 and 81.
  - (i) If  $\gcd(a, b) = 1$  and  $\gcd(a, c) = 1$ , then  $\gcd(a, bc) = 1$ .
  - (j)  $\gcd(a, b) = \gcd(b, a \bmod b)$ . 1 x 10 = 10 marks.
2. Suppose that  $S$  is a set of  $n$  integers. Show that one can always find a nonempty subset  $T$  of  $S$  such that the sum of all the elements of  $T$  is divisible by  $n$ .  
Example: Let  $n = 4$  and  $S = \{3, 7, 21, -5\}$ . Since  $(3 + 21)$  is divisible by 4, we can take  $T = \{3, 21\}$ . 10 marks.

3. Let  $\mathcal{F}$  be a family (or collection) of  $m$  subsets of a finite set  $X$ . For any  $x \in X$ , let  $p(x)$  be the number of pairs  $(A, B)$  of sets  $A, B \in \mathcal{F}$  such that either  $x \in A \cap B$  or  $x \notin A \cup B$ . Prove that  $p(x) \geq \frac{m^2}{2}$ , for every  $x \in X$ . (You may use the fact that for any two positive real numbers  $a$  and  $b$ ,  $a^2 + b^2 \geq \frac{(a+b)^2}{2}$ . )
- Example: Let  $X = \{1, 2, 3\}$ ,  $\mathcal{F} = \{\{1, 2\}, \{1\}, \{2\}\}$ . So, here  $m = |\mathcal{F}| = 3$  and  $\frac{m^2}{2} = 4.5$ . Let  $A := \{1, 2\}$ ,  $B := \{1\}$ ,  $C := \{2\}$ . Then,  $p(1) = |\{(A, A), (A, B), (B, A), (B, B), (C, C)\}| = 5 \geq 4.5$ . Similarly, one can see that both  $p(2)$  and  $p(3)$  are at least 4.5. 10 marks.

————— ALL THE BEST —————